

GOVERNMENT OF TAMIL NADU
Department of School Education



TN-SCHOOLS

Unified Educational Ecosystem

Comprehensive User Guide · Technical Portal Documentation

- Version 2.0
- Government Schools
- Private Schools
- 20+ AI Content Tools
- State Executive Telemetry

VERSION	SCOPE	CLASSIFICATION
2.0 — Production Reference	All Role Portals Across State Hierarchy	Administrative, Teaching, Student & Governance



Confidentiality & Operational Handbook Notice

This document serves as the complete technical user manual and operational handbook for all administrative, teaching, student, parent, and governance portals operating within the Tamil Nadu School Education System. Access and permissions are strictly enforced according to state Role-Based Access Control (RBAC) protocols.



Unified Digital Educational Architecture

The TN-Schools ecosystem bridges state-level policy formulation with day-to-day classroom execution. By interconnecting distinct user roles across schools and education departments, it automates attendance telemetry, student academic mastery tracking, welfare disbursements, teacher pedagogy with 20+ specialized AI tools, and executive state intelligence.

TABLE OF CONTENTS

1. Application Overview & Strategic Purpose	
1.1 Core Strategic Objectives	
1.2 Educational Governance Hierarchy	
1.3 System Portal Directory	
1.4 Private vs Government School Modules	
1.5 Technology Stack & Architecture	
2. Authentication & Access Control	
2.1 How to Login & Role-Based Redirection	
2.2 Security & Data Isolation Principles	
3. Headmaster Portal	
3.1 Dashboard & Administrative Cockpit	
3.2 Student Management & Enrollment	
3.3 Staff & Faculty Management	
3.4 Parent Directory & Records	
3.5 Attendance Overview & Verification	
3.6 Master Timetable Management	
3.7 Exams & Results Board	
3.8 Scholarships & Government Schemes	
3.9 Mid-Day Meal (MDM) Daily Audits	
3.10 Alumni, Clubs & Events	
3.11 Digital Library & Resources	
3.12 Promotions & Academic Year Rollover	
3.13 Rewards & Recognition	
3.14 School Public Portal & Gallery	

4. Teacher Portal

- 4.1 Dashboard & Classroom Command
- 4.2 Attendance Management
- 4.3 Homework & Assignments
- 4.4 AI Lesson Planner
- 4.5 AI Evaluation & Vision OCR Grading
- 4.6 Question Bank & Generator
- 4.7 Timetable & Proxy Classes
- 4.8 Student Profiles & Progress
- 4.9 Exams & Assessment Marking
- 4.10 Parent Communication & PTA Slots
- 4.11 Digital Library & Study Materials
- 4.12 Science Labs & Virtual Experiments
- 4.13 NEET & SSLC Board Preparation
- 4.14 Social Activities & School Press
- 4.15 Sports, Cultural Events & Clubs
- 4.16 Analytics & Early-Warning Risk Alerts
- 4.17 AI Content Studio (20 Specialized Tools)

5. PET — Physical Education Teacher Portal

- 5.1 Dashboard
- 5.2 Student Fitness Records & BMI
- 5.3 Sports Events & Tournament Board
- 5.4 Awards, Medals & Sports Quota
- 5.5 Sports Equipment Inventory Ledger
- 5.6 Injury Reports & Ground Maintenance
- 5.7 Sports Clubs & Athletic Teams

6. Student Portal

- 6.1 Differentiated Grade Hubs (Classes 6–12)
- 6.2 Homework & Assessments
- 6.3 Learning Hub & 24/7 AI Tutor
- 6.4 Science & Virtual Lab Features
- 6.5 Digital Student Portfolio & Activities
- 6.6 Exam Preparation (SSLC / NEET)
- 6.7 Health, Wellness & Safety Desk

7. Parent Portal

- 7.1 Dashboard & Ward Overview
- 7.2 Daily Attendance & Absence Alerts
- 7.3 Academic Performance & Term Reports
- 7.4 Homework & Assignment Monitor

7.5 Direct Communication with Teachers	
7.6 PTA Appointments & Assemblies	
7.7 Leave Applications & Approvals	
7.8 Health & Physical Growth Records	
7.9 Scholarship & Entitlements Tracker	
7.10 Notifications & Circulars	
7.11 AI Parent Assistant	
7.12 Child's Verified Digital Portfolio	
8. Government Officer Portals	
8.1 BEO — Block Education Officer Portal	
8.2 DEO — District Education Officer Portal	
8.3 Commissioner of School Education Portal	
8.4 Minister Dashboard (Executive Command Center)	
9. Cross-Portal Workflows & Operational Lifecycles	
9.1 Daily Attendance & Early-Risk Intervention Lifecycle	
9.2 Pedagogical Cycle: AI Planning to Automated Grading	
9.3 Mid-Day Meal (MDM) Transparency Lifecycle	
9.4 Academic Year Rollover & Promotion Lifecycle	
9.5 School Press & Public Broadcast Lifecycle	
10. Notifications & Alerts System	
10.1 Automated Real-Time Triggers	
10.2 Notification Delivery Architecture	
11. Master Roles & Portal Applicability Matrix	
11.1 Master Roles & Supervised Entities Quick Reference	
11.2 Private vs Government Feature Matrix	



Chapter 1: Application Overview & Strategic Purpose

1.1 Core Strategic Objectives

The **TN-Schools Unified Educational Platform** is an enterprise-scale digital governance and personalized learning environment developed to streamline school management across Tamil Nadu's 38 revenue districts. The application establishes a transparent, automated, and AI-enabled data highway connecting all tiers of the education hierarchy.



Equitable Learning Access

Delivers centralized Samacheer Kalvi digital textbooks, interactive virtual science and vocational laboratories, and bilingual AI tutoring to every student regardless of geographic location.



Pedagogical AI Empowerment

Equips educators with an AI Content Studio featuring 20 specialized tools for automated lesson planning, Bloom's taxonomy question generation, and OCR-assisted grading.



Real-Time Telemetry

Provides immediate visibility into daily student and staff attendance, Mid-Day Meal (MDM) grain stocks, teacher deployments, and infrastructure health.



Dropout Risk Elimination

Leverages predictive tracking algorithms to identify irregular attendance patterns early and initiate community counseling workflows.



Family & School Partnership

Gives parents transparent, real-time insights into their ward's academic standing, health indices, and direct two-way messaging with educators.

1.2 Educational Governance Hierarchy

The ecosystem strictly mirrors the administrative governance structure of the Tamil Nadu Department of School Education:

Hierarchy Level	Primary Authority	Scope of Administrative Oversight
State Apex Executive	Minister for School Education	State-wide policy direction, live telemetry, public grievances, and legislative intelligence.
State Operational Directorate	Commissioner of School Education	State operations, budget utilization, inter-district analytics, and policy compliance.
District Administration	District Education Officer (DEO)	District school rankings, dropout heatmaps, teacher rationalization, and capital infrastructure.

Hierarchy Level	Primary Authority	Scope of Administrative Oversight
Sub-District Administration	Block Education Officer (BEO)	Block school inspections, headmaster supervision, mid-day meal audits, and counselor dispatch.
Institutional Leadership	Headmaster / Principal	School-wide staff lifecycle, student enrollment, promotions, assets, and government schemes.
Instructional Execution	Teachers & PET Staff	Classroom teaching, daily attendance, AI-driven evaluation, sports fitness, and parent communication.
Beneficiaries & Community	Students & Parents	Self-paced learning, virtual labs, test preparation, attendance visibility, assessments, and welfare tracking.

1.3 System Portal Directory

Portal	Target Persona	Key Operational Focus
Student Portal	Students (Grades 6–12)	AI Tutor, self-study plans, mock tests, virtual labs, and student portfolio.
Parent Portal	Parents & Guardians	Child performance, attendance alerts, health log, and direct teacher chat.
Teacher Portal	Class & Subject Teachers	AI lesson planner, OCR evaluation, question bank, and attendance logging.
PET Portal	Physical Education Staff	Student fitness records, sports events, inventory, and ground audits.
Headmaster Portal	Principals & Headmasters	Staff lifecycle, student admissions, mid-day meal, and school resources.
BEO Portal	Block Education Officers	Block school monitoring, teacher deployment, and dropout radars.
DEO Portal	District Education Officers	District school rankings, dropout geospatial heatmaps, and learning outcomes.
Commissioner Portal	Commissioner of Education	State operations, budget utilization, and cross-district analytics.
Minister Dashboard	Minister for Education	Live executive command center, AI predictive policy, and media briefs.

1.4 Private vs Government School Modules

The platform seamlessly adapts features based on the school's administrative category:

- **Core Modules (Common to All):** Student Management, Daily Attendance, Staff Directory, Timetable, Exams & Results, Digital Library, Parent Communication, Student Portfolios, AI Lesson Planning.
- **Government School Specializations:** Mid-Day Meal (MDM) Daily Audits, State Welfare Distribution (Free Uniforms, Laptops, Bicycles, Textbooks), Scholarship Approvals, BEO Promotion Approvals, Staff Transfer

Requests.

- **Optional Advanced Add-ons:** Specialized Science Labs (Physics, Chemistry, Botany, Zoology), NEET & SSLC Board Boost Modules, Interactive Infographics Hub.

1.5 Technology Stack & Architecture

Next.js 14 & React

High-performance modern frontend architecture with responsive layouts and server-side rendering.

Node.js & Express

Robust backend REST API handling multi-tenant data pipelines and role permissions.

AI Studio Engines

Integrated vision OCR for handwritten paper grading, bilingual AI tutor, and Bloom's taxonomy generator.

RBAC & JWT Security

Strict role-based tokens ensuring zero cross-tenant leakage between schools and districts.

Chapter 2: Authentication & Access Control

2.1 How to Login & Role-Based Redirection

Users access the system through a unified, secure login interface. Upon entering their credentials (EMIS ID / Username and Password), the authentication server verifies their credentials and role, immediately redirecting them to their dedicated portal cockpit.

User Role	Primary Identifier	Landing Dashboard Route
Headmaster	School HM Account / EMIS Code	/headmaster
Teacher	Teacher Staff ID	/teacher
PET Staff	PET Staff ID	/pet
Student	Student Roll / EMIS Number	/student
Parent	Registered Mobile / OTP	/parent
BEO Officer	Block Admin Credentials	/block-education-officer
DEO Officer	District Admin Credentials	/district-education-officer
Commissioner	State Directorate Account	/commissioner
Minister	Executive State ID	/minister

2.2 Security & Data Isolation Principles

The TN-Schools platform enforces complete cryptographic school-level data isolation. A school's staff, students, and marks cannot be accessed by any other school. State and district officers access aggregated analytical metrics while protecting individual student privacy.



Chapter 3: Headmaster Portal

The **Headmaster Portal** serves as the primary operational cockpit for managing institutional compliance, faculty administration, student lifecycle, and state welfare schemes.

3.1 Dashboard & Administrative Cockpit

Provides a bird's-eye view of today's attendance summary, staff attendance, pending leave approvals, mid-day meal counts, and upcoming school events.

3.2 Student Management & Enrollment

Enables individual student admissions, bulk Excel uploads, section assignments, Transfer Certificate (TC) generation, and student registry exports.

3.3 Staff & Faculty Management

Complete teaching, contract, and temporary staff directory, service registers, leave approval workflows, and substitute/proxy timetable assignments.

3.4 Parent Directory & Records

Directory of parent contacts linked with student EMIS IDs, communication history, and PTA membership records.

3.5 Attendance Overview & Verification

Real-time institutional attendance compilation, absentee tracking, and daily submission confirmation to BEO/DEO offices.

3.6 Master Timetable Management

Class-wise and teacher-wise timetable builder, period conflict detection, and proxy period allocation for absent teachers.

3.7 Exams & Results Board

Examination schedules, hall ticket distribution, model exam analytics, and top performer rankings across classes.

3.8 Scholarships & Government Schemes

Distribution logs for government welfare schemes (uniforms, textbooks, footwear, bicycles, laptops) and state scholarship approvals.

3.9 Mid-Day Meal (MDM) Daily Audits

Daily headcount logging of meals served, grain inventory reconciliations (rice, dhal, eggs), nutritional quality checks, and kitchen hygiene compliance.

3.10 Alumni, Clubs & Events

Verified alumni network, co-curricular club management (Eco Club, Red Cross, NCC, NSS), and annual celebration logs.

3.11 Digital Library & Resources

Catalog of physical and digitized library books, teacher resource repositories, and digital content repository management.

3.12 Promotions & Academic Year Rollover

End-of-year batch student promotions from class to class, retention logs, and promotion list submission to BEO for government approvals.

3.13 Rewards & Recognition

Student achievement badges, teacher appreciation entries, and athletic awards tracking.

3.14 School Public Portal & Gallery

Configuration of the school's public website profile, photo galleries, staff showcase, and public announcement board.

Chapter 4: Teacher Portal

The **Teacher Portal** combines daily classroom administrative management with state-of-the-art AI tools to elevate pedagogical quality and cut administrative workload by up to 60%.

4.1 Dashboard & Classroom Command

Displays today's schedule, assigned classes, quick attendance status, pending homework to grade, and recent student risk alerts.

4.2 Attendance Management

Fast single-tap attendance marking for assigned classrooms, automatic parent SMS trigger for absentees, and monthly attendance percentage records.

4.3 Homework & Assignments

Create assignments with attachments, set submission deadlines, view student submission rates, and grade submitted files.

4.4 AI Lesson Planner

Automated lesson blueprint generator producing Samacheer Kalvi-compliant plans complete with teaching objectives, timeline splits, and interactive classroom activities.

4.5 AI Evaluation & Vision OCR Grading

Vision-enabled AI scanning tool that reads handwritten student answer scripts, suggests score allocations against custom rubrics, and generates qualitative student feedback.

4.6 Question Bank & Generator

Generates multi-tier questions (Easy, Medium, Hard) mapped to Bloom's taxonomy, producing instant question papers with answer keys.

4.7 Timetable & Proxy Classes

View weekly assigned periods, room numbers, and accept proxy substitution classes when assigned by the Headmaster.

4.8 Student Profiles & Progress

Comprehensive 360-degree student profile showing attendance trends, exam marks, co-curricular achievements, and teacher remarks.

4.9 Exams & Assessment Marking

Input term and unit test marks, generate class grade distributions, and automatically publish report cards to the Parent Portal.

4.10 Parent Communication & PTA Slots

Two-way messaging with parents, broadcast classroom announcements, and schedule PTA meeting slots.

4.11 Digital Library & Study Materials

Upload study notes, presentations, question banks, and reference links accessible by students in the Learning Hub.

4.12 Science Labs & Virtual Experiments

Interactive virtual laboratory simulations in Physics, Chemistry, Botany, and Zoology for classroom demonstrations.

4.13 NEET & SSLC Board Preparation

Specialized revision modules, previous year question paper banks, and timed mock tests for Class 10 and 12 board readiness.

4.14 Social Activities & School Press

Document community service projects, social awareness rallies, and submit articles for the school press newsletter.

4.15 Sports, Cultural Events & Clubs

Nominate students for inter-house cultural competitions, drama clubs, and coordinate with the Physical Education Teacher.

4.16 Analytics & Early-Warning Risk Alerts

Automated early-warning engine flagging students showing chronic absenteeism or sharp performance drops, initiating timely academic remediation.

4.17 AI Content Studio (20 Specialized Tools)

Suite of 20 subject-adaptive pedagogical tools organized into four key teaching domains:

Pedagogical Domain	Available AI Tools in Studio	Teaching Impact
Teach & Explain	Concept Explainer, Analogy Maker, Real-World Storyboard, Visual Mindmap Generator, Bilingual Tamil-English Translator	Simplifies complex scientific & mathematical concepts with relatable regional analogies.
Practice & Work	Worksheet Builder, Step-by-Step Math Solver, Vocabulary Flashcard Maker, Diagram Drawing Prompter, Lab Protocol Assistant	Produces customized classroom exercises and guided problem-solving sheets.
Tests & Assess	Bloom's Taxonomy Question Creator, Rubric Architect, Multiple Choice Generator, Short Answer Evaluator, Model Answer Key Maker	Generates tiered assessment items covering Knowledge, Understanding, and Application.

Pedagogical Domain	Available AI Tools in Studio	Teaching Impact
Differentiate	Remedial Lesson Builder, Gifted Student Challenge Generator, Special Needs Simplifier, English Language Scaffold, At-Risk Study Booster	Personalizes classroom materials to match every student's learning pace.



Chapter 5: PET — Physical Education Teacher Portal

The **PET Portal** provides a dedicated infrastructure for tracking student physical health indices, athletic development, school sports inventory, and zonal competition entries.

5.1 Dashboard

Summary of upcoming sports meets, ground readiness status, daily fitness record counts, and athlete rosters.

5.2 Student Fitness Records & BMI

Records height, weight, BMI, endurance tests, 100m sprint timings, and motor skill assessments across all classes.

5.3 Sports Events & Tournament Board

Management hub for annual sports meets, zonal athletic fixtures, inter-school matches, and team registrations.

5.4 Awards, Medals & Sports Quota

Digital registry documenting medals, certificates, state athletic representations, and sports quota verification records.

5.5 Sports Equipment Inventory Ledger

Asset log tracking balls, gymnastics gear, athletic kits, damage reports, and new procurement requisitions.

5.6 Injury Reports & Ground Maintenance

Safety and maintenance audit logs for running tracks, playing courts, football fields, and sports injury first-aid logs.

5.7 Sports Clubs & Athletic Teams

Team rosters for Cricket, Football, Volleyball, Kabaddi, Kho-Kho, and Track & Field squads.



Chapter 6: Student Portal

The **Student Portal** provides an engaging, personalized digital learning hub designed to support self-directed study, test readiness, and career exploration.

6.1 Differentiated Grade Hubs (Classes 6–12)

The student portal automatically adapts its interface, learning tools, and navigation to the student's grade level:



Middle School Hub (Classes 6–8)

Gamified learning with achievement badges, interactive storybooks, math formula flashcards, and curiosity challenges.



High School Hub (Classes 9–10)

SSLC Board examination preparation cockpit, previous year question papers (PYQs), timed mock tests, and score predictors.



Higher Secondary Specialized Streams (Classes 11–12)

Dynamic stream adaptation for Science, Commerce, Computer Science, and Arts & Vocational streams with dedicated virtual labs and competitive prep.

6.2 Homework & Assessments

View pending homework, submission deadlines, submit completed files/photos, and check teacher marks with feedback.

6.3 Learning Hub & 24/7 AI Tutor

Bilingual (Tamil/English) AI assistant providing step-by-step guidance on complex homework problems, chapter summaries, and interactive lessons.

6.4 Science & Virtual Lab Features

Virtual science experiments, 3D biological models, chemical reaction simulations, and interactive science infographics.

6.5 Digital Student Portfolio & Activities

Verified digital repository of academic certificates, extracurricular badges, sports achievements, and leadership records.

6.6 Exam Preparation (SSLC / NEET)

Mock test engine, syllabus breakdown, revision planner, formula cheat sheets, and model exam rankings.

6.7 Health, Wellness & Safety Desk

Personal health card, wellness tips, leave application requests, and confidential counselor communication channel.



Chapter 7: Parent Portal

The **Parent Portal** fosters active family engagement by providing comprehensive transparency regarding child progress, institutional notifications, and welfare entitlements.

7.1 Dashboard & Ward Overview

Overview of the child's daily attendance status, overall academic grade, recent test results, pending homework, and teacher remarks.

7.2 Daily Attendance & Absence Alerts

Live attendance calendar tracking daily check-ins, excused leaves, and automated SMS alerts triggered upon unexcused morning absence.

7.3 Academic Performance & Term Reports

Subject-wise marks breakdown, term report cards, teacher feedback notes, and progress mastery charts.

7.4 Homework & Assignment Monitor

Real-time ledger of pending, submitted, and evaluated homework assignments with submission deadline reminders.

7.5 Direct Communication with Teachers

Secure messaging channel to communicate directly with class and subject teachers regarding academic or behavioral progress.

7.6 PTA Appointments & Assemblies

Book one-on-one appointments for Parent-Teacher Association (PTA) meetings and view assembly notifications.

7.7 Leave Applications & Approvals

Submit digital leave applications for sickness or personal reasons directly to the class teacher, receiving instant status updates.

7.8 Health & Physical Growth Records

Annual physical examination records, vision screening test reports, height/weight BMI growth curves, and vaccination tracking.

7.9 Scholarship & Entitlements Tracker

Status tracking for government assistance including free uniforms, textbooks, bicycles, laptops, and direct scholarship bank transfers.

7.10 Notifications & Circulars

Official school circulars, holiday announcements, examination timetables, and emergency alerts.

7.11 AI Parent Assistant

Intelligent conversational assistant offering guidance on supporting child study routines, understanding exam schedules, and navigating welfare schemes.

7.12 Child's Verified Digital Portfolio

View verified co-curricular certificates, athletic awards, and sign parent reflections for school records.



Chapter 8: Government Officer Portals

Comprehensive multi-tier governance portals designed for administrative oversight across Tamil Nadu's educational hierarchy.

8.1 BEO — Block Education Officer Portal

Target Users: Sub-District Block Education Officers overseeing 50–100 schools in an educational block.

- **Block School Registry & Inspections:** Master registry of all government, aided, and private schools with historical inspection logs and audit reports.
- **Headmaster Administration:** Administrative circular broadcasting, compliance tracking, and HM meetings.
- **Teacher Deployment & PTR Rationalization:** Pupil-Teacher Ratio (PTR) analysis across the block, identifying staffing deficits and deploying temporary substitute teachers.
- **Dropout Prevention Radar:** Block-wide early-warning radar tracking students with prolonged absenteeism, enabling rapid dispatch of field counselors.

- **Mid-Day Meal Audits:** Block-level grain distribution audits, kitchen hygiene compliance tracking, and nutrition anomaly flags.

8.2 DEO — District Education Officer Portal

Target Users: District Education Officers (DEOs) and Chief Educational Officers (CEOs) across Tamil Nadu's 38 revenue districts.

- **District School Rankings:** Composite index ranking schools across academic pass percentages, student attendance consistency, and athletic achievements.
- **Geospatial Dropout Heatmap:** Geographic visualization pinpointing high-risk dropout pockets across urban, rural, and tribal sectors.
- **Block Governance & Coordination:** Comparative evaluation of block performance metrics, inspection completions, and administrative throughput.
- **Infrastructure Capital Projects:** Monitoring school building construction, smart classroom rollouts, science lab modernizations, and sanitation facilities.
- **Learning Outcomes Benchmarking:** Multi-year analysis of Foundational Literacy and Numeracy (FLN) and board exam pass trends.

8.3 Commissioner of School Education Portal

Target Users: Commissioner of School Education, Joint Directors, and State Directorate Leadership.

- **State Operational Telemetry:** Live state summary of active student enrollments, daily teacher attendance rates, and composite education indices across all 38 districts.
- **DEO Performance Scorecards:** Evaluation of district leadership across KPI benchmarks, inspection completion rates, and scheme execution timelines.
- **Budget Utilization & Grant Tracking:** Real-time fiscal monitoring of Samagra Shiksha grants, infrastructure disbursements, and teacher development funds.
- **Statewide Policy Enforcement:** Tracking rollout compliance for updated Samacheer Kalvi curricula, teacher training milestones, and digital lab deployments.

8.4 Minister Dashboard (Executive Command Center)

Target Users: Honourable Minister for School Education, Principal Secretaries, and Cabinet Advisors.

- **Live State Telemetry Cockpit:** Real-time executive dashboard monitoring daily attendance check-ins, breakfast/lunch meal disbursements, and operational status across all districts.
- **AI Policy & Performance Predictions:** Predictive modeling engines forecasting 5-year enrollment trends, board exam pass forecasts, and demographic fiscal requirements.
- **Flagship Welfare Schemes Progress:** Real-time disbursement metrics for flagship initiatives such as the Chief Minister's Breakfast Scheme, Free Laptops, Pudhumai Penn, and Illam Thedi Kalvi.
- **Public Grievance & Media Intelligence:** Media sentiment synthesis, public feedback digests, and legislative briefing generators for assembly questions.



Chapter 9: Cross-Portal Workflows & Operational Lifecycles

The TN-Schools platform connects multiple roles into seamless automated workflows. The operational lifecycles below illustrate how data and responsibility move across the system.

9.1 Daily Attendance & Early-Risk Intervention Lifecycle

Attendance Telemetry & Dropout Prevention Cycle

- 1 Classroom Check-In:** The class teacher marks digital attendance through the Teacher Portal by 9:30 AM.
- 2 Instant Parent Notification:** Absence records trigger immediate automated SMS and push notifications to parents through the Parent Portal.
- 3 School-Wide Aggregation:** The Headmaster Portal compiles morning totals and verifies institutional attendance rates.
- 4 Block & District Telemetry:** BEO and DEO portals highlight schools with anomalous absentee spikes.
- 5 Watchlist Intervention:** If a student accumulates five or more unexcused absences, the system automatically creates an At-Risk case, alerting the Headmaster and dispatching a field counselor.

9.2 Pedagogical Cycle: AI Planning to Automated Grading

Classroom AI Pedagogy & Evaluation Flow

- 1 AI Lesson Formulation:** The teacher generates a Samacheer Kalvi-aligned lesson plan through the Teacher Portal's lesson planner.
- 2 Adaptive Question Generation:** The teacher uses the Question Generator to produce tiered assessment questions mapped to Bloom's taxonomy.
- 3 Student Practice & Assessment:** Students attempt practice quizzes on the Student Portal or complete written classroom sheets.
- 4 Vision AI OCR Evaluation:** The teacher scans physical answer sheets using the AI Evaluation tool, which performs OCR mark suggestions and rubric grading.
- 5 Performance Analytics Distribution:** Evaluated marks automatically populate Student and Parent dashboards, highlighting specific conceptual weaknesses for remediation.

9.3 Mid-Day Meal (MDM) Transparency Lifecycle

Mid-Day Meal & Welfare Verification Pipeline

- 1 **Institutional Meal Count:** The Headmaster records morning student meal counts and grain consumption through the Headmaster Portal.
- 2 **Block-Level Quota Audit:** The BEO monitors block grain stock balances and flags supply deficits before stocks deplete.
- 3 **District Hygiene & Nutrition Log:** The DEO inspects meal quality logs and verifies random inspection reports.
- 4 **State Executive Dashboard:** Commissioner and Minister dashboards reflect live statewide nutrition metrics, verifying budget utilization in real time.

9.4 Academic Year Rollover & Promotion Lifecycle

Grade Promotion & Institutional Rollover Pipeline

- 1 **Year-End Marks Verification:** Teachers finalize annual term scores and generate promotion eligibility lists.
- 2 **Headmaster Verification:** Headmaster audits promotion records and submits government school batches for BEO endorsement.
- 3 **BEO Formal Approval:** Block Officer reviews and authorizes promotion rollouts across government schools in the block.
- 4 **Automated Rollover:** The system rolls students to the next academic grade, archiving previous year transcripts in their permanent digital portfolio.

9.5 School Press & Public Broadcast Lifecycle

Public Announcement & Media Approval Flow

- 1 **Draft Submission:** Teachers and student editorial clubs draft achievement stories and photo galleries.
- 2 **Headmaster Editorial Review:** Headmaster reviews and approves the news item for public release.
- 3 **Multi-Channel Publication:** Approved news instantly publishes to the school's public website, parent notifications, and student activity feeds.



Chapter 10: Notifications & Alerts System

10.1 Automated Real-Time Triggers

Event Trigger	Recipient	Notification Channel	Content & Action Required
Morning Absence	Parent	SMS & In-App Alert	Immediate alert that child is marked absent by 9:30 AM. Parent can reply or submit leave.
Homework Assigned	Student & Parent	In-App Notification	Subject, topic, submission due date, and attached digital worksheets.
Exam Marks Published	Student & Parent	In-App & SMS	Term / unit test marks published with score breakdown and teacher remarks.
5+ Absentee Alert	Headmaster & BEO	High-Priority Alert	At-risk student flagged for dropout prevention counselor intervention.
Low MDM Grain Stock	Headmaster & BEO	System Warning	Alert that rice/dhal inventory is below 3-day buffer threshold.
PTA Meeting Request	Parent / Teacher	In-App Invitation	Appointment slot confirmation for parent-teacher conference.

10.2 Notification Delivery Architecture

The notification system utilizes real-time WebSockets for instant in-app alerts and an enterprise SMS gateway for mission-critical parent absence notifications, ensuring zero missed alerts even in rural areas with low internet connectivity.



Chapter 11: Master Roles & Portal Applicability Matrix

11.1 Master Roles & Supervised Entities Quick Reference

A consolidated lookup of every role, its span of supervision, and its core daily actions within the platform:

Role	Direct Supervised Entities	Core Daily Actions
Student	Self (Academics & Portfolio)	Attend AI tutoring, take mock tests, complete assignments, access virtual labs.
Parent	Linked Wards / Children	Review attendance alerts, check test marks, message teachers, track welfare entitlements.
Teacher	Assigned Classrooms & Students	Mark attendance, generate AI lesson plans, evaluate tests with OCR, manage homework.

Role	Direct Supervised Entities	Core Daily Actions
PET Staff	Athletics & Sports Inventory	Record student fitness tests, manage tournaments, track sports equipment and ground safety.
Headmaster	Single School Staff & Students	Approve staff leaves, audit mid-day meals, oversee admissions, timetable, and school operations.
BEO	50–100 Schools in Block	Conduct school inspections, balance teacher deployment, audit MDM, track dropouts.
DEO	All Blocks & High Schools in District	Rank district schools, audit dropout heatmaps, monitor infrastructure projects, FLN trends.
Commissioner	All 38 Districts & State Departments	Execute state education policies, track budget utilization, review DEO performance scorecards.
Minister	State Executive Command	Review live statewide telemetry, evaluate AI policy forecasts, review media briefs, welfare schemes.

11.2 Private vs Government Feature Matrix

Feature / Module	Private Schools	Government Schools	State Governance Portals
Student Admissions & Attendance	✔ Enabled	✔ Enabled	👁 Telemetry View
Teacher AI Studio (20 Tools)	✔ Enabled	✔ Enabled	—
Vision AI OCR Answer Grading	✔ Enabled	✔ Enabled	—
24/7 Bilingual AI Student Tutor	✔ Enabled	✔ Enabled	—
Mid-Day Meal (MDM) Daily Audits	—	✔ Enabled	👁 State Audits
State Welfare Schemes (Laptops/Bicycles)	—	✔ Enabled	👁 Budget & Tracking
BEO Promotion Approvals	Internal Approval	✔ Required	✔ Block Authorization
Dropout Early-Warning Radar	Internal Alerts	✔ Enabled	✔ District Counselor Dispatch
Virtual Science Laboratories	✔ Enabled	✔ Enabled	—
Public School Web Portal & Gallery	✔ Enabled	✔ Enabled	—



Official Verification & Technical Support

For technical support, role credential provisioning, or syllabus integration requests, contact the State IT Support Desk through the official portal support channel or by email at support@tnschools.gov.in.